

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY- CHANGA

B.Tech. EE Sem - VI Examination, May 2014
EE - 307 Electrical Power System -III

Date: 08/05/2014, Thursday

Time: 10.00 a.m. to 1.00 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

- Q-1**
- i. A 3-phase circuit breaker is rated at 2000 MVA, 33KV. Its making current will be... [7]
(a) 35 kA (b) 70 kA (c) 89 kA (d) 161 kA
 - ii. Which of the following sequences of operation represents the rated operating duty cycle of a circuit breaker? (O-open; C-close; t=3s; T=3 minutes)
(a) O-t-CO-T-CO (b) O-CO-t-CO-T-CO
(c) O-C-t-OC-T (d) O-CO-T-CO-T-C
 - iii. The resistance of an electric arc can be increased by
(a) Increasing the concentration of ionized particles.
(b) reducing the arc length,
(c) Splitting the arc. (d) Increasing the arc x-section.
 - iv. In a 220 KV system, the inductance and capacitance up to the circuit breaker location are 25 mH and 0.025 μ F respectively. The value of resistor required to be connected across the breaker contacts which will give no transient oscillations is
(a) 25 Ω (b) 250 Ω (c) 500 Ω (d) 1000 Ω
 - v. It is difficult to interrupt a capacitive circuit because
(a) the current has a leading power factor
(b) the restriking voltage can be high
(c) current magnitude is very small
(d) stored energy in the capacitor is very high
 - vi. During arc extinction SF₆ gas gets
(a) decomposed into SF₄ and SF₂ (b) reduced to SF₂
(c) decomposed into S and F ions (d) oxidised
 - vii. The arc voltage in a circuit breaker is
(a) in the phase with the arc current (b) lagging the arc current by 90°
(c) leading the arc current by 90° (d) lagging the arc current by 180°
- Q-2**
- A. Explain the effect of power factor and armature reaction on Transient recovery voltage. [6]
 - B. Explain the different causes due to which deionization of gases take place. [3]
 - C. With neat diagrams explain the term “Current Chopping”. [5]

OR

- Q-2 A. Derive an expression for the restriking voltage in terms of system voltage, inductance and capacitance across a circuit breaker contact. [6]
 B. Write short technical notes on (i) kilometric fault, (ii) phase opposition switching. [5]
 C. In a short circuit test on a circuit breaker, the following readings obtained on a frequency transient: (a) time to reach peak restriking voltage 70 μ sec. (b) the peak restriking voltage 100 KV. Calculate the average rate of rise of restriking voltage and the natural frequency of the circuit. [3]
- Q-3 A. Explain the "High resistance arc interruption" with relevant diagrams. [7]
 B. Write short note on selection of circuit breaker [7]

OR

- Q-3 A. With neat diagram explain the construction and working of vacuum circuit breaker. [7]
 B. List out the physical and dielectric properties of SF₆ gas. [7]

SECTION – II

- Q -4 (i) The power invariance theorem says that... [7]
 (a) In all sequence network power remains same.
 (b) Zero sequence network doesn't consume any power.
 (c) Power consume as per phase quantity is three times power consumes as per positive sequence quantity.
 (d) Power consume as per phase quantity is equal to sum of the power consume in positive, negative and zero sequence quantity.
- (ii) A part of power system is shown in figure 1. For which combination of transformer connection, the L-L fault current value will be equal to L-L-G fault current.

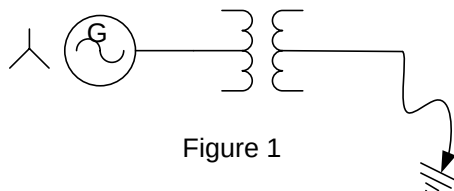


Figure 1

- (a) Delta/ Star Grounded
 (b) Star Grounded/ Delta
 (c) Star Grounded/ Star Grounded
 (d) For any combination, L-L fault will never equal to L-L-G Fault current
- (iii) For the L-L fault at shown location of following figure 2, as value of X_n increases, the effect on fault current is.....

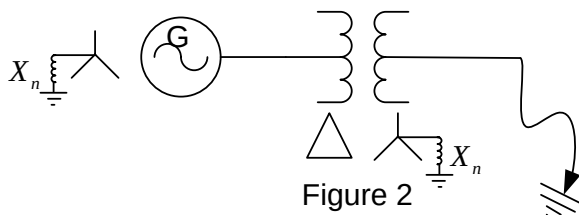


Figure 2

- (a) Fault current increases (b) Fault current decrease
 (c) Fault current remains same (d) No fault current will flow

(iv) For the sequence current, which statement is true during normal condition as well as during fault condition?

- (a) $I_2 > 100\%$ of I_1
- (b) $I_2 < 20\%$ of I_1
- (c) $I_2 \leq 50\%$ of I_1
- (d) $I_2 \leq 100\%$ of I_1

(v) A part of power system is shown in figure 3. When the switch 'S' is in open condition, at that time a single line to ground fault occurs at shown location of transmission line. From following options which option is correct?

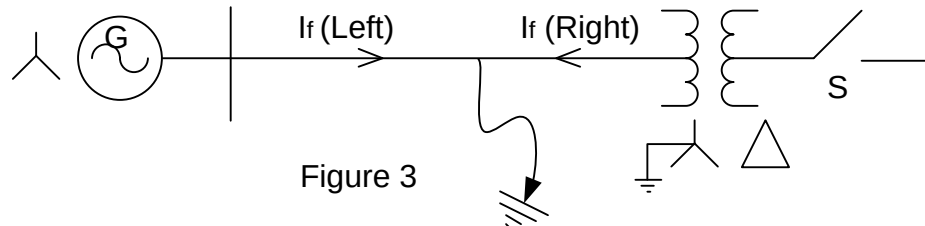


Figure 3

- (a) $I_f (\text{Left})=0, I_f (\text{Right})=0$
- (b) $I_f (\text{Left})=0, I_f (\text{Right})$ will flow
- (c) $I_f (\text{Left})$ will flow, $I_f (\text{Right})=0$
- (d) $I_f (\text{Left})$ will flow, $I_f (\text{Right})$ will flow
- (e) Data is insufficient

(vi) Series reactors are used to

- (a) improve the transmission efficiency.
- (b) improve the power factor of the power system.
- (c) improve the voltage regulation.
- (d) bring down the fault level within the capacity of the switchgear.

(vii) For a particular fault, $I_{c0}=0$ and $I_{b2}=0$. The possible fault is_____.

- (a) L-G (b) L-L (c) L-L-G (d) L-L-L

Q-5 (A) Derive the equation of transients on transmission line. Also prove the doubling [7] effect.

(B) A simple power system is shown in figure 4. When the motors were running at [7] unity p.f. with 80% load with motor bus voltage of 12 KV, at that time three phase fault occur on generator bus. Find the fault current. Neglect the prefault current and **take base as per generator rating.**

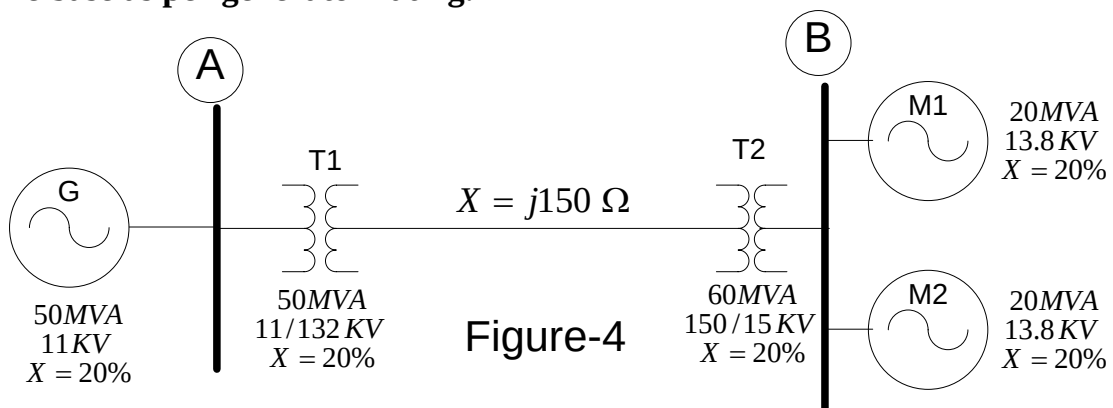
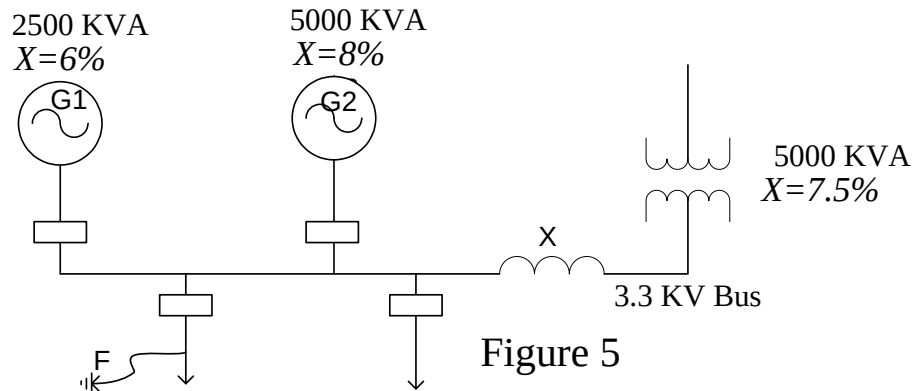


Figure-4

OR

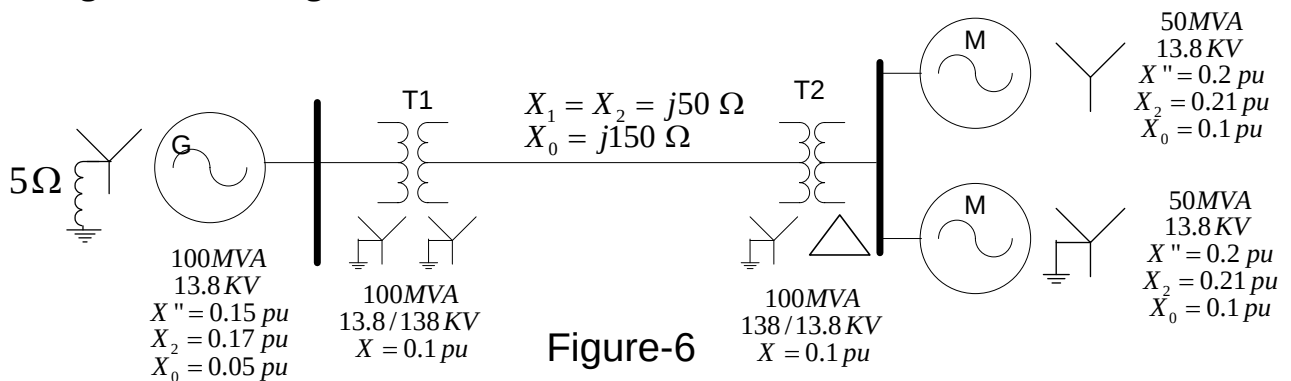
Q-5 (A) Explain in detail: the steady state and short circuit model of synchronous machine. [7]

(B) A power-station has two alternators of 2,500 KVA and 5,000 KVA capacities and the reactances are 6 % and 8 % respectively. The circuit breakers are rated 150 MVA. The system is to be extended by taking supply from a grid via transformer of 5,000 KVA rating having a reactance of 7.5 %. If the same circuit breakers are to be used, find the necessary reactance to be placed between the bus bar section of the station and the supply from the grid to protect the switchgear. The voltage of the system at the bus-bars is 3-phase, 3.3 kV. The layout is shown in figure 5. **Take base KVA=5000 KVA** [7]



Q-6 (A) Using appropriate interconnection of sequence networks, derive the equation for a double line fault in a power system with a fault impedance of Z_f . [7]

(B) A part of power system is shown in figure 6. Find value of LLG fault current in ampere when fault occurs at the middle of transmission line. **Take the base as per generator rating.** [7]



OR

Q-6 (A) Prove the power invariance theorem in connection with sequence components. [7]

(B) A part of power system is shown in **figure 6**. Find value of LG fault current when fault occurs at the motor terminal with fault resistance of 1 ohm. **Take the base as per generator rating.** [7]
